Abhi Patel

abhip202@gmail.com | (647)-741-8735 | b33boy | abhi-patel-0 | abhipatel.netlify.app

EDUCATION

Bachelor of Engineering (Honours) - Mechatronics Engineering, Sep 2017 – Apr 2022 | Oshawa, Canada
Ontario Tech University (Graduated President's list with highest distinction)

WORK EXPERIENCE

Software Engineer (NASA's Lunar Gateway - Canadarm3), May 2022 – Mar 2024 | Brampton, Canada
Macdonald, Dettwiler & Associates (MDA) - Robotics & Space Ops Division

- Played a key role in developing **mission-critical flight and ground software**, applying strong engineering principles and **SDLC/ATDD methodologies** to ensure reliability, traceability, and system resilience in a concurrent environment.
- Developed **C++ modules compliant with NASA software standards**, integrating telemetry data from auto-generated embedded code for real-time arm and joint state monitoring.
- Architected a **persistent telemetry logging system** in C# (.NET) using **SOLID principles and Object-Oriented Design**, enabling **reliable data recovery and analysis**.
- Served as **Quality Lead**, driving **testing best practices** and ensuring **acceptance tests aligned with customer requirements** to improve test coverage and delivery confidence.
- Created Python automation scripts for document generation and test verification, **saving hours of manual work per release and improving consistency across test cycles**.
- Leveraged **Builder and Factory design patterns** to modularize message transmission, data mapping, and client request handling systems, **reducing future module development time by 50%** and enhancing code maintainability.
- Built a React-based interface for creating and dispatching flight commands (e.g. lock, limp, move-joints), **streamlining command workflows and improving operator efficiency in mission-critical environments**.

Cloud Development Contractor (Telus Insights), Jan 2020 – Sep 2020 | Scarborough, Canada
Telus Communications - Systems For Enterprise Analytics

- Leveraged Terraform to **automate GCP infrastructure** as part of the Telus Insights Project, leading to **$1,600 in weekly cost savings** by removal of redundant VMs.
- Collaborated with **cross-functional teams** including external vendors, offshore QA, and product owners to **deliver infrastructure milestones on time and within budget**, ensuring alignment between architecture and business goals.
- Conducted in-depth analysis of **GCP Compute Engine instance pricing tiers**, selecting configurations tailored to project memory and performance requirements achieving **$2,000+ in monthly savings** through informed architectural trade-offs.

Engineering Co-op (Telus Insights), May 2019 – Dec 2019 | Scarborough, Canada
Telus Communications - Systems For Enterprise Analytics

- Developed shell scripts within GCP Compute Engine to dynamically populate large volumes of data into **Cloud Storage Buckets**, enabling faster ingestion into BigQuery and saving the team **~10 hours per week**.
- Identified and resolved a major bottleneck in an Apache Airflow data ingestion pipeline, **accelerating data availability and enabling faster downstream processing.**
- Created **Python automation scripts** to extract data from internal sources, reducing task time to **under 30 seconds.**
- Integrated **CI/CD** into multiple Git repositories using Cloud Build, **shortening feedback cycles**.

PROJECTS

Redis Clone, *High-Performance Key-Value Store*

- Built a **Redis-inspired key-value store in C++**, achieving **<3ms latency and 20k+ requests per second**, leveraging **I/O multiplexing** for efficient handling of concurrent TCP connections without multithreading.
- Utilized raw **socket programming** for fine control over networking and applied **CRTP to reduce virtual dispatch overhead**, optimizing performance.
- Employed **modern C++ patterns (RAII, Rule of Five, move semantics, etc)** and performance profiling to achieve scalable, efficient code.

Content Discoverability Engine for Youtube Videos

- Developed a **containerized microservice backend in Go and Python** to automatically ingest, process, and index video captions for enhanced search and discoverability. Deployed to **GCP** using **Terraform** for consistent and repeatable operations.
- Designed and integrated a third-party API for data acquisition, transformation, and storage. Managed data integrity through a dual-write pattern to **MySQL** and **Elasticsearch**.

Vision Based 3D Scanner, *Capstone Project*

- Developed an image processing pipeline for isolating line laser data and **optimized it to minimize sensitivity to environmental lighting** using OpenCV.
- Implemented **pose estimation** with ChArUco fiducial markers to **achieve accurate 3D positioning** of the scanner.
- Generated a **point cloud** representation from the final scan data for **3D visualization**.

Autonomous Package Recovery Robot

- Programmed a Turtlebot with a ROS node (explore_lite) to perform **frontier exploration via Simultaneous Localization and Mapping (SLAM)** within a 25m² room, returning to its starting pose.
- Developed a ROS node (C++/Python) to implement a **full-coverage path planning algorithm**, using marker detection to locate and retrieve two boxes in a timely manner.
- Visualized generated maps in RViz and conducted simulations in Gazebo, **significantly streamlining the testing process and reducing the need for manual intervention**.

Autonomous Maze Navigating Robot

- **Successfully led** a group to build a **differential drive robot that can autonomously navigate a two-level maze** and pick up a small objects.
- Performed electrical analysis and **produced schematics** for efficient power regulation as well as transmission of signals to onboard electronics (i.e., sensors, motor driver, and microprocessor).
- Programmed and tuned a **wall-following algorithm** that implemented **PID control** to determine motor speeds given **distance sensor** inputs, allowing the robot to drive in a straight line.
- Performed tests on, and **calibrated encoders and distance sensors** to produce meaningful and accurate measurements.

Mimix - Conversational Mimic Bot

- Designed and built a Discord chatbot that mimics the user's messages using a local LLM and **vector similarity search** via **ChromaDB**, enabling personalized, context-aware conversations

AWARDS

Undergraduate Student Research Award, May 2021
Natural Sciences and Engineering Research Council of Canada

- Implemented a **convolutional neural network** to detect cracks in power lines.
- Performed **data augmentation** to generate a larger, more diverse dataset of images.

SKILLS

Languages — C++, C, Python, Go, C#, JavaScript, TypeScript | **Technologies** — Docker, Kubernetes, Terraform, GCP, AWS, MySQL, MongoDB, RabbitMQ, ROS, OpenCV, Valgrind, perf, gdb, PyTorch

CERTIFICATES

- Machine Learning (Coursera)
- Deep Learning Specialization (Coursera)
- Object Oriented Data Structures in C++ (Coursera)
- Docker Mastery (Udemy)